

Environmental Sound Classification Using Temporal and Spectral Audio Features

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GitHub project: [HTTPS://GITHUB.COM/IRISHE177/ELEC5305-PROJECT-540662177](https://github.com/IrisHe177/ELEC5305-PROJECT-540662177)

1. Project objective and progress

This work aims to investigate how temporal and spectral audio features can distinguish environmental sounds. The current work provides a reproducible ESC-10 baseline and tests if simple yet interpretable descriptors improve classification when added to MFCCs. Data preparation, feature extraction, five-fold training and evaluation, result tables, confusion matrices and exploratory sound pairs analysis have been completed.

2. Dataset and evaluation protocol

The experiment uses the predefined ESC-10 subset of ESC-50: 400 five-second recordings for 10 balanced classes with 40 clips per class[2]. The classes are dog, rooster, rain, sea waves, crackling fire, crying baby, sneezing, clock tick, helicopter and chainsaw.

Item	Current implementation
Audio	44,100 Hz; mono; 5 seconds per clip
Evaluation	Five official folds; 320 training and 80 test clips per fold
Split checks	Eight test clips per class per fold; source recordings do not cross folds
Preprocessing	Original audio retained; no amplitude normalization
Feature scaling	Training-fold means and population standard deviations only

3. Feature extraction and comparison

Analysis uses 2,048-sample frames (46.44 ms), a 512-sample hop (11.61 ms), a periodic Hann window and a 2,048-point FFT for spectral features. Reflection padding produces 431 analysis frames per clip. Each feature is summarized by its mean and population standard deviation. The MFCC + ZCR baseline and clip-level statistical aggregation are adapted from Piczak [2]. MFCCs use twelve coefficients excluding C0; RMS is calculated before windowing. ZCR uses 512-sample chunks at the same hop. Traditional audio descriptors such as ZCR, spectral centroid and rolloff have been used in audio classification [1]

Spectral descriptors use one-sided FFT power; rolloff is the frequency containing 90% of that power.

Zero-spectrum frames receive zero-valued spectral descriptors. All groups use the same official folds and fixed model settings.

Group	Features	Dimensions
A: baseline	MFCC + ZCR	26
B: descriptors	RMS + ZCR + centroid + spread + flatness + rolloff	12
C: combined	MFCC + all six descriptors in B; ZCR appears once	36

4. Quantitative results

The SVM uses a Gaussian kernel, $C = 1$, $\text{KernelScale} = \text{sqrt}(26)$, and one-versus-one classification for every feature group. No parameters are tuned. A five-neighbour Euclidean classifier with equal weights provides an additional comparison. Group A produces identical predictions to the separately implemented baseline in this project.

SVM group	Accuracy	Macro P	Macro R	Macro F1	Fold SD
A (26)	64.75	65.65	64.75	64.84	7.42
B (12)	60.75	61.67	60.75	60.44	2.88
C (36)	73.50	74.60	73.50	73.70	4.37

Table 1. SVM results. Metrics are percentages; fold SD is in percentage points.

kNN group	Accuracy	Macro P	Macro R	Macro F1
A	57.25	57.52	57.25	55.44
B	56.75	56.81	56.75	55.93
C	61.75	62.85	61.75	60.34

Table 2. kNN results, reported using the same out-of-fold evaluation.

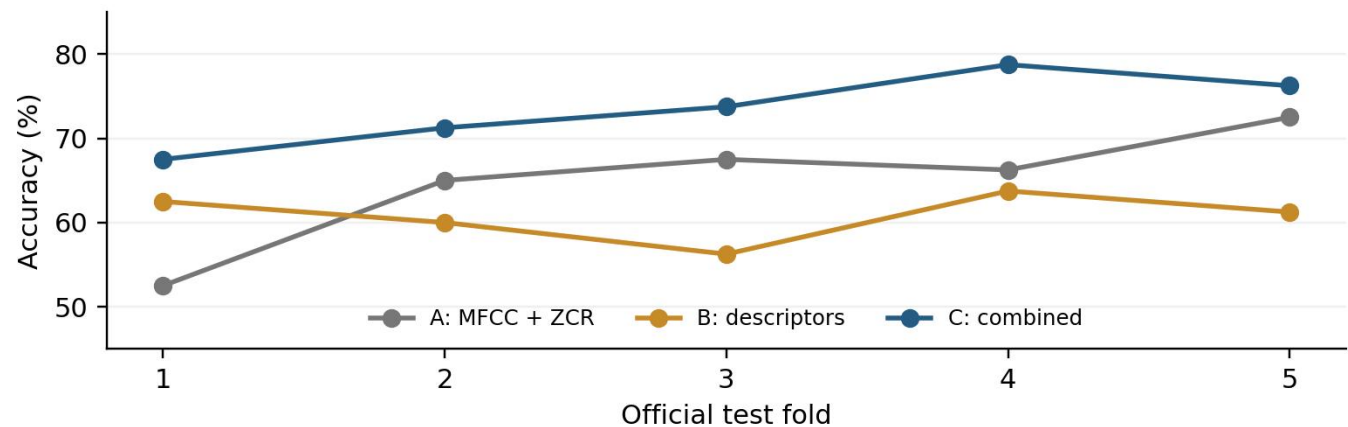


Figure 1. SVM accuracy on each official test fold.

4.1 Interpretation

Group C has the highest accuracy (73.50%) and macro-F1 (73.70%) and outperforms A by 8.75 and 8.86%, respectively. It corrects 49 baseline errors and adds 14, which adds 35 more correct predictions to the set, and outperforms A in all five folds. Group B has 60.75% accuracy below the baseline, but the combination of the feature improvement suggests complementary information. This is in agreement with Chu et al. [5], but their features differ. kNN accuracy also increases by 4.50 percentage points with C.

Accuracy is calculated from 400 pooled out-of-fold predictions. Macro precision, recall and F1 are unweighted averages of class-level metrics. Macro-recall equals accuracy because the classes are balanced. Fold SD describes variation between five fold accuracies and is not a confidence interval.

5. Confusion and feature-distribution analysis

Sea waves and chainsaw are selected because they have the largest total bidirectional confusion in the baseline SVM. This selection describes the baseline errors; it is not the most confused pair in every feature group.

SVM group	Sea waves → chainsaw	Chainsaw → sea waves	Total
A	8	13	21
B	2	8	10
C	4	3	7

Table 3. Misclassifications between the selected categories, pooled over the official folds.

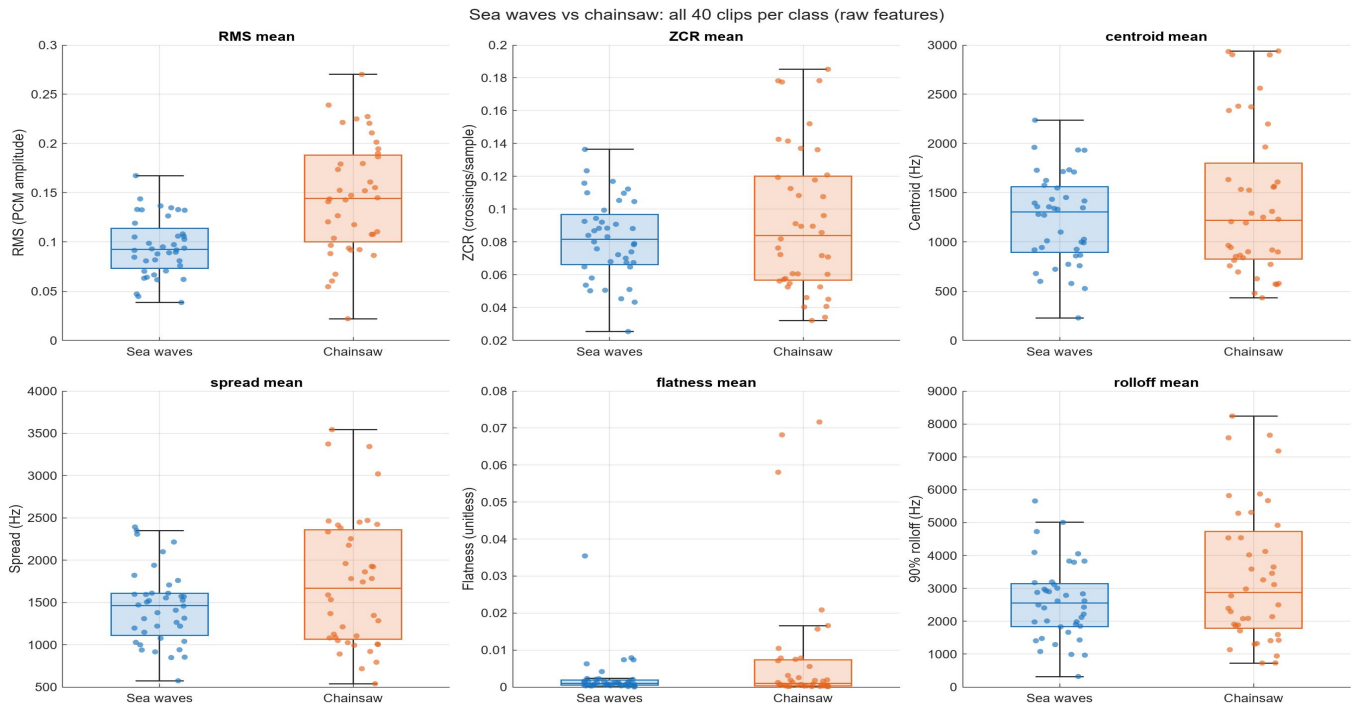


Figure 2. Raw clip-level feature means for all 40 sea-wave and 40 chainsaw recordings. These distributions are exploratory descriptions, not scaled model inputs.

5.1 What the distributions suggest

The chosen pair is still difficult to choose, as many of the descriptors overlap. Median ZCR is around 0.0817 for sea waves and 0.0839 for chainsaw. The median spectral flatness is also similar (about 0.000945 and 0.000995). These summaries alone do not account for much separation between the two groups. Chainsaw has a higher median frame-RMS mean (0.1443 vs. 0.0925) and spectral spread (1,666 vs. 1,464 Hz) and rolloff (about 2,878 vs. 2,556 Hz) than sea waves. However, the distributions are similar; centroid medians are about 1,220 Hz for chainsaw and 1,307 Hz for sea waves. There is no single threshold to separate the classes.

Combining the features reduces the pair total from 21 to 7 errors. This supports further investigation of complementary descriptors but does not establish which descriptor is causing the improvement. There is also confusion elsewhere; group C has 15 mutual errors between crackling fire and clock tick.

6. Example recordings and limitations

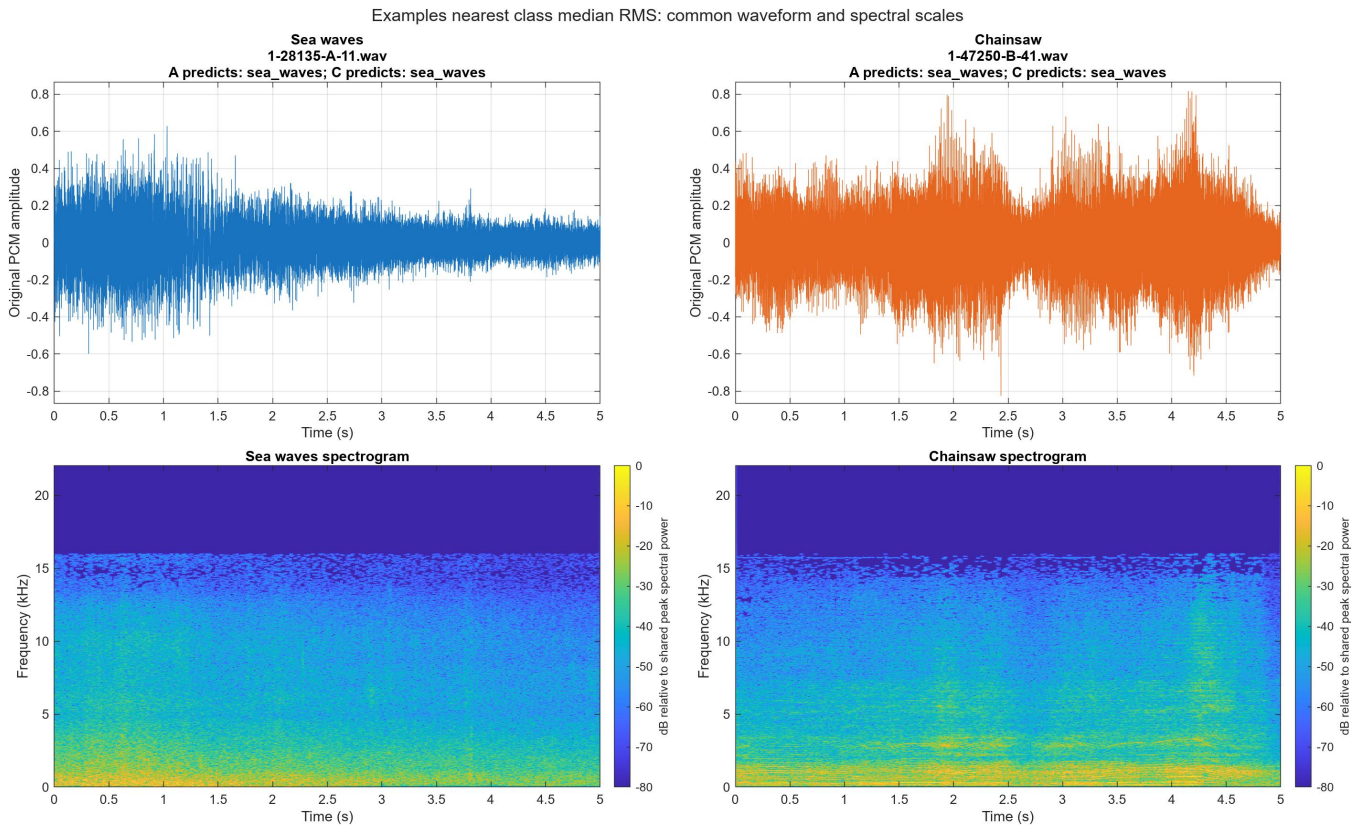


Figure 3. Example waveforms and spectrograms. Files: 1-28135-A-11.wav (sea waves) and 1-47250-B-41.wav (chainsaw). Both plots use a shared spectral reference and a -80 to 0 dB colour range.

Examples are selected near each class's median RMS, independently of prediction correctness. Both A and C identify sea waves correctly but misclassify chainsaw. Both examples show broadband energy, yet cannot establish class-wide behaviour. This preliminary comparison uses fixed parameters and the same kernel scale across different feature dimensions, which may affect performance. Gains do not

establish statistical significance or optimal accuracy. Mean/std summaries discard temporal order, while RMS may reflect recording gain or microphone distance.

7. Next steps and reproducibility

Next work will analyse class-level errors, assess descriptor contributions through ablation, and explore temporal or delta features. Hyperparameters will be selected within the training data, keeping official test folds separate and the evaluation protocol consistent.

MATLAB R2024b saves features, predictions, metrics and confusion matrices. Current results come from `run_20261009_031441_478` and its sound analysis. Code and progress are available through the GitHub link on page 1.

Sources: [ESC-50 DATASET](#) and saved project outputs. Results remain preliminary.

References Identified to Date

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- [3] J. Salamon, C. Jacoby, and J. P. Bello, “A dataset and taxonomy for urban sound research,” in *Proceedings of the 22nd ACM International Conference on Multimedia*, 2014, pp. 1041–1044, doi: 10.1145/2647868.2655045.
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